Project Report

MTH210: Statistical Computing

Name: Naina Bhalla **Roll Number:** 240674

1 Problem Statement

The dataset has been generated from the following PDF with $\alpha > 0$ and $\lambda > 0$.

$$f(x; \alpha, \lambda) = \alpha\lambda e^{-\lambda x}(1 - e^{-\lambda x})^{\alpha-1} \quad \text{for } x > 0 \tag{1}$$

and 0, otherwise. Find the MLEs of α and λ based on Newton Raphson (NR) algorithm and also calculate 95% confidence intervals of both α and λ based on parametric and non-parametric bootstrap methods.

2 Derivations

To implement the Newton-Raphson (NR) algorithm, we first derive the log-likelihood function, its gradient, and the Hessian matrix.

Likelihood Function For a sample $\{X_1, X_2, ...X_n\}$ of size n from $f(x; \alpha, \lambda)$,

$$L(\alpha, \lambda \mid \underline{x}) = \alpha^n \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \prod_{i=1}^n (1 - e^{-\lambda x_i})^{\alpha-1} \tag{2}$$

Log-Likelihood Function The log-likelihood function $l(\alpha, \lambda)$ is given by:

$$l(\alpha, \lambda \mid \underline{x}) = n\ln(\alpha) + n\ln(\lambda) - \lambda \sum_{i=1}^n x_i + (\alpha - 1)\sum_{i=1}^n \ln(1 - e^{-\lambda x_i}) \tag{3}$$

First Derivatives Taking the partial derivatives with respect to α and λ:

$$\frac{\partial l}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^n \ln(1 - e^{-\lambda x_i}) \tag{4}$$

$$\frac{\partial l}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i + (\alpha - 1)\sum_{i=1}^n \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}} \tag{5}$$

Second Derivatives (Hessian Matrix) The elements of the Hessian matrix H are:

$$\frac{\partial^2 l}{\partial \alpha^2} = -\frac{n}{\alpha^2} \tag{6}$$

$$\frac{\partial^2 l}{\partial \alpha \partial \lambda} = \sum_{i=1}^{n} \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}} = \frac{\partial^2 l}{\partial \lambda \partial \alpha} \tag{7}$$

$$\frac{\partial^2 l}{\partial \lambda^2} = -\frac{n}{\lambda^2} - (\alpha - 1) \sum_{i=1}^{n} \frac{x_i^2 e^{-\lambda x_i}}{(1 - e^{-\lambda x_i})^2} \tag{8}$$

Therefore, the Hessian matrix is:

$$H = \begin{bmatrix} \frac{\partial^2 l}{\partial \alpha^2} & \frac{\partial^2 l}{\partial \alpha \partial \lambda} \\ \frac{\partial^2 l}{\partial \lambda \partial \alpha} & \frac{\partial^2 l}{\partial \lambda^2} \end{bmatrix} \tag{9}$$

Newton-Raphson The Newton-Raphson update step is computed iteratively as:

$$\begin{bmatrix} \alpha_{k+1} \\ \lambda_{k+1} \end{bmatrix} = \begin{bmatrix} \alpha_k \\ \lambda_k \end{bmatrix} - H^{-1} \begin{bmatrix} \frac{\partial l}{\partial \alpha} \\ \frac{\partial l}{\partial \lambda} \end{bmatrix} \tag{10}$$

The convergence criterion used is the L_∞ norm (maximum absolute difference):

$$\max\left(|\alpha_{k+1} - \alpha_k|, |\lambda_{k+1} - \lambda_k|\right) < \text{tol} \tag{11}$$

3 Approach

3.1 Initialization for Newton-Raphson

To ensure rapid and stable convergence, a data-driven initialization approach was utilized.

- **Initial Guess for λ (λ_0):** By temporarily assuming $\alpha = 1$, the Generalized Exponential (GE) distribution simplifies to a standard Exponential distribution. The closed-form MLE for a standard Exponential distribution is simply the inverse of the sample mean. Thus, $\lambda_0 = 1/\bar{x}$ provides a decent starting value.

- **Initial Guess for α (α_0):** By substituting λ_0 into the equation 4 for α and setting it to zero ($\frac{\partial l}{\partial \alpha} = 0$), we isolate α. This yields $\alpha_0 = -n / \sum_{i=1}^{n} \ln(1 - e^{-\lambda_0 x_i})$.

3.2 Computational Safeguards

- **Enforcing Domain Positivity:** The support of the distribution requires strictly $\alpha > 0$ and $\lambda > 0$. At each iteration, any parameter update that becomes negative is truncated to a computationally safe minimum bound of 10^{-6} using the max() function.

- **Singular Matrix Protection:** During non-parametric bootstrapping, heavily skewed random resamples can cause the Hessian matrix to become computationally singular and impossible to invert. To prevent algorithmic failure, the optimization calls were wrapped in a `try()` block to silently catch singular matrix errors, discard the failed resample, and proceed without crashing the entire simulation loop.

3.3 Bootstrap

To calculate 95% confidence intervals for the parameter vector $\theta = (\alpha, \lambda)$, both nonparametric and parametric bootstrap methods ($B = 1000$ replications) were implemented.

3.3.1 Nonparametric Bootstrap

In nonparametric bootstrap, we resample data of size n from the empirical distribution of the observed data $\{X_1, X_2, \ldots, X_n\}$ with replacement, and obtain estimates of θ using these samples. Let $\hat{\theta}_b^*$ denote the MLE from the b-th bootstrap sample:

$$\begin{gathered} X_{11}^*, X_{21}^*, \ldots, X_{n1}^* \Rightarrow \hat{\theta}_1^* \\ X_{12}^*, X_{22}^*, \ldots, X_{n2}^* \Rightarrow \hat{\theta}_2^* \\ \vdots \\ X_{1B}^*, X_{2B}^*, \ldots, X_{nB}^* \Rightarrow \hat{\theta}_B^* \end{gathered}$$

The B estimates $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ form an approximate sample from the sampling distribution of $\hat{\theta}$. By ordering these bootstrap estimates such that $\hat{\theta}_{(1)}^* < \hat{\theta}_{(2)}^* < \cdots < \hat{\theta}_{(B)}^*$, a $100(1-\alpha)\%$ confidence interval is constructed by taking the $(\alpha/2)$-th and $(1-\alpha/2)$-th quantiles. For a 95% interval ($\alpha = 0.05$), the limits are:

$$(L, U) = \left(\hat{\theta}_{\lfloor 0.025 \cdot B \rfloor}^*, \hat{\theta}_{\lfloor 0.975 \cdot B \rfloor}^*\right)$$

3.3.2 Parametric Bootstrap

Suppose the original data follows the GE distribution $F(x; \alpha, \lambda)$, and let $\hat{\theta} = (\hat{\alpha}, \hat{\lambda})$ be the initial MLEs. Instead of resampling the observed data, the parametric bootstrap generates new samples of size n directly from the estimated parametric model $F(\hat{\theta})$:

$$\begin{gathered} X_{11}^*, X_{21}^*, \ldots, X_{n1}^* \sim F(\hat{\theta}) \Rightarrow \hat{\theta}_1^* \\ X_{12}^*, X_{22}^*, \ldots, X_{n2}^* \sim F(\hat{\theta}) \Rightarrow \hat{\theta}_2^* \\ \vdots \\ X_{1B}^*, X_{2B}^*, \ldots, X_{nB}^* \sim F(\hat{\theta}) \Rightarrow \hat{\theta}_B^* \end{gathered}$$

This method assumes the GE distribution model is correct. Using the empirical MLEs ($\hat{\alpha}$ and $\hat{\lambda}$), new datasets are generated strictly from the given PDF $f(x)$. Samples were generated via **Inverse Transformation**. By equating the CDF to a uniform random variable $U \sim \text{Unif}(0, 1)$ and solving for x:

$$x = \frac{-\ln(1 - U^{1/\hat{\alpha}})}{\hat{\lambda}} \tag{12}$$

As with the nonparametric method, the 95% confidence interval is determined by the 2.5th and 97.5th percentiles of the resulting ordered bootstrap estimates.

4 Questions

1. **File Number:** The data set was retrieved from file number **fort.22**.

2. **Initial Guess Values:** For the initial guess of λ, the inverse of the sample mean ($\lambda_0 = 1/\bar{x}$) was used, which is the MLE of λ for a standard exponential distribution (This can be obtained by setting $\alpha = 1$). For the initial guess of α, the first derivative equation of α was equated to zero and substituted with λ_0, yielding $\alpha_0 = -n/\sum_{i=1}^{n} \ln(1 - e^{-\lambda_0 x_i})$.

3. **Stopping Criterion:** The algorithm utilized the L_∞ norm (maximum absolute error) of the parameter update step. Convergence was achieved when the maximum absolute difference between consecutive parameter estimates fell below a tolerance of 10^{-6} (i.e., $\max(|\alpha_{k+1} - \alpha_k|, |\lambda_{k+1} - \lambda_k|) < 10^{-6}$).

4. **Iterations to Converge:** The 2D Newton-Raphson algorithm converged in **4** iterations.

5. **Bootstrap Replications:** For both the parametric and non-parametric bootstrap confidence intervals, $B = 1000$ replications were used.

5 Results

The computed Maximum Likelihood Estimates and their corresponding 95% confidence intervals are summarized in Table 1.
The Newton-Raphson converged in 4 iterations. All the results below are calculated after using **set.seed(0)** for reproducibility.

Table 1: MLEs and 95% Confidence Intervals

Parameter	**Estimate ($\hat{\theta}$)**	**95% CI (Parametric)**	**95% CI (Non-Parametric)**
α	1.192037	(0.7756832, 2.367928)	(0.8223298, 2.324108)
λ	0.6255646	(0.3929639, 1.042976)	(0.409288, 1.093849)